PALAK BHATI

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Portfolio

Professional Summary

Aspiring **AI/ML engineer** and BCA student specializing in **Generative AI**, **retrieval-augmented generation**, and **NLP**. Hands-on with **LangChain**, **Sentence-Transformers**, **Qdrant**, and **Groq-hosted LLMs**, building **RAG pipelines** with source-cited answers, access-controlled retrieval, and automated evaluation through **Ragas** and **LangSmith**. Applies classical **ML** alongside **LLMs**, including an **embedding-based classifier** and **DBSCAN clustering** used to cut unnecessary model calls. Seeking **AI/ML** or **Generative AI internship** roles.

Skills

Programming: Python, SQL, JavaScript, HTML, CSS, DSA
Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Hugging Face Transformers
Frameworks: LangChain, LangGraph, LlamaIndex, FastAPI, Flask, Streamlit
AI/ML: Machine Learning basics, NLP, Fine-tuning, Model Evaluation
Generative AI: LLMs, RAG, Prompting, Embeddings, Vector Databases, Document Loaders, Text Splitters, Retrievers
Agentic AI: AI Agents, Agentic RAG, Prompt Chaining, Routing, Parallelization, Orchestrator-Worker, Evaluator-Optimizer, Human-in-the-Loop (HITL), Memory and Persistence, Streaming, MCP
LLM Platforms: OpenAI GPT, Claude, Gemini, Llama, Groq, Ollama
Tools: Git, GitHub, Docker, Linux, LangSmith, VS Code, Jupyter Notebook, Google Colab
Cloud/Deployment: Google Cloud, Render

Projects

Secure-RAG-RBAC-Chatbot GitHub | Live

- Built a secure **RAG chatbot** using **LangChain**, **Groq LLMs**, **Qdrant vector database**, and **Sentence-Transformers**, with **Docling** document ingestion, header-aware chunking, and source-cited answers in a **Streamlit UI**.
- Implemented **role-based access control** for 5 user roles using native **Qdrant payload filtering** at retrieval time, preventing unauthorized document access. Validated it with **audit logging** and an 11-case **Pytest permission suite**.
- Developed layered **LLM guardrails** combining regex and LLM-based **prompt injection detection** with output-side **PII redaction**, role-aware masking of sensitive HR data, and **groundedness checks** to reduce hallucinations.
- Automated **RAG evaluation** with **Ragas** in a **GitHub Actions CI pipeline** that blocks regressions. Integrated **LangSmith tracing**, per-request LLM cost tracking, and **Docker deployment**.

Hybrid NLP Log Classification System GitHub | Live

- Engineered a **3-tier log classification pipeline** combining regex rules, an **embedding-based ML model**, and an **LLM fallback**, routing logs by source to balance cost and accuracy.
- Trained a **Logistic Regression classifier** on **all-MiniLM-L6-v2 embeddings**, achieving **99 percent accuracy** and **0.98 macro F1** on held-out data, with confidence thresholding for uncertain predictions.
- Applied **DBSCAN clustering** on sentence embeddings of **2,400+ logs** to discover recurring patterns and convert them into regex rules, reducing load on the ML and LLM layers.
- Integrated a **Groq-hosted LLM** with structured prompting for low-data log sources and deployed the pipeline as a **Dockerized FastAPI**

Certifications and Achievements

- Active **Open-source contributor** with merged PRs in Github repositories.
- **150+ Data Structures and Algorithms** problems solved on LeetCode.
- Hands-on with **Claude Code** for agentic, AI-assisted software development workflows.
- Hands-on with **Google Cloud Platform (GCP)** services for cloud deployment.
- Completed Udemy Certifications in **AI Engineering**, **Google Cloud Platform (GCP)**, and **Python**.

Education

- **Holkar Science College, Indore** 2024 – 2027
  Bachelor of Computer Application (BCA)